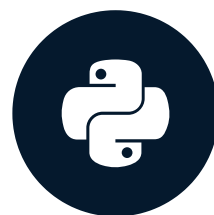


# Arrays

INTRODUCTION TO PYTHON FOR FINANCE



**Adina Howe**  
Instructor

# Installing packages

```
pip3 install package_name_here
```

```
pip3 install numpy
```

# Importing packages

```
import numpy
```

# NumPy and Arrays

```
import numpy  
my_array = numpy.array([0, 1, 2, 3, 4])  
print(my_array)
```

```
[0, 1, 2, 3, 4]
```

```
print(type(my_array))
```

```
<class 'numpy.ndarray'>
```

# Using an alias

```
import package_name  
package_name.function_name(...)
```

```
import numpy as np  
my_array = np.array([0, 1, 2, 3, 4])  
print(my_array)
```

```
[0, 1, 2, 3, 4]
```

# Why use an array for financial analysis?

- Arrays can handle very large datasets efficiently
  - Computationally-memory efficient
  - Faster calculations and analysis than lists
  - Diverse functionality (many functions in Python packages)

# What's the difference?

## NumPy arrays

```
my_array = np.array([3, 'is', True])  
print(my_array)
```

```
['3' 'is' 'True']
```

## Lists

```
my_list = [3, 'is', True]  
print(my_list)
```

```
[3, 'is', True]
```



# Array operations

## Arrays

```
import numpy as np

array_A = np.array([1, 2, 3])
array_B = np.array([4, 5, 6])

print(array_A + array_B)
```

```
[5 7 9]
```

## Lists

```
list_A = [1, 2, 3]
list_B = [4, 5, 6]

print(list_A + list_B)
```

```
[1, 2, 3, 4, 5, 6]
```



# Array indexing

```
import numpy as np
```

```
months_array = np.array(['Jan', 'Feb', 'March', 'Apr', 'May'])  
print(months_array[3])
```

```
Apr
```

```
print(months_array[2:5])
```

```
['March' 'Apr' 'May']
```

# Array slicing with steps

```
import numpy as np
```

```
months_array = np.array(['Jan', 'Feb', 'March', 'Apr', 'May'])
```

```
print(months_array[0:5:2])
```

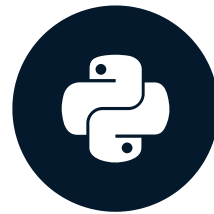
```
['Jan' 'March' 'May']
```

# Let's practice!

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# Two Dimensional Arrays

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# Two-dimensional arrays

```
import numpy as np
```

```
months = [1, 2, 3]
```

```
prices = [238.11, 237.81, 238.91]
```

```
cpi_array = np.array([months, prices])
```

```
print(cpi_array)
```

```
[[ 1.    2.    3. ]  
 [238.11 237.81 238.91]]
```

# Array Methods

```
print(cpi_array)
```

```
[[ 1.    2.    3. ]  
 [238.11 237.81 238.91]]
```

`.shape` gives you dimensions of the array

```
print(cpi_array.shape)
```

```
(2, 3)
```

`.size` gives you total number of elements in the array

```
print(cpi_array.size)
```

```
6
```

# Array Functions

```
import numpy as np
```

```
prices = [238.11, 237.81, 238.91]  
prices_array = np.array(prices)
```

`np.mean()` calculates the mean of an input

```
print(np.mean(prices_array))
```

```
238.27666666666667
```

`np.std()` calculates the standard deviation of an input

```
print(np.std(prices_array))
```

```
0.46427960923946671
```



# The `arange()` function

`numpy.arange()` creates an array with start, end, step

```
import numpy as np
```

```
months = np.arange(1, 13)  
print(months)
```

```
[ 1  2  3  4  5  6  7  8  9 10 11 12]
```

```
months_odd = np.arange(1, 13, 2)  
print(months_odd)
```

```
[ 1  3  5  7  9 11]
```



# The `transpose()` function

`numpy.transpose()` switches rows and columns of a numpy array

```
print(cpi_array)
```

```
[[ 1.    2.    3. ]  
 [238.11 237.81 238.91]]
```

```
cpi_transposed = np.transpose(cpi_array)
```

```
print(cpi_transposed)
```

```
[[ 1. 238.11]  
 [ 2. 237.81]  
 [ 3. 238.91]]
```

# Array Indexing for 2D arrays

```
print(cpi_array)
```

```
[[ 1.    2.    3. ]  
 [238.11 237.81 238.91]]
```

```
# row index 1, column index 2  
cpi_array[1, 2]
```

```
238.91
```

```
# all row slice, third column  
print(cpi_array[:, 2])
```

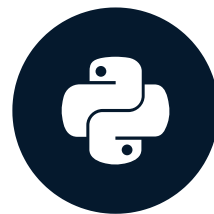
```
[ 3.    238.91]
```

# Let's practice!

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# Using Arrays for Analyses

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# Indexing Arrays

```
import numpy as np
```

```
months_array = np.array(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'])  
indexing_array = np.array([1, 3, 5])
```

```
months_subset = months_array[indexing_array]  
print(months_subset)
```

```
['Feb' 'Apr' 'Jun']
```

# More on indexing arrays

```
import numpy as np

months_array = np.array(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'])

negative_index = np.array([-1, -2])

print(months_array[negative_index])
```

```
['Jun' 'May']
```

# Boolean arrays

```
import numpy as np

months_array = np.array(['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'])

boolean_array = np.array([True, True, True, False, False, False])

print(months_array[boolean_array])
```

```
['Jan' 'Feb' 'Mar']
```



# More on Boolean arrays

```
prices_array = np.array([238.11, 237.81, 238.91])  
# Create a Boolean array  
boolean_array = (prices_array > 238)  
  
print(boolean_array)
```

```
[ True False  True]
```

```
print(prices_array[boolean_array])
```

```
[ 238.11  238.91]
```



# Let's practice!

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